

Mechanisms of Convective Initiation: A Literature Review

Purpose. This review catalogs the mechanisms that the literature has theoretically and observationally linked to the initiation of deep moist convection (CI), organized so they can be mapped onto quantities **observable from AIRS soundings and SMAP soil moisture** (with HYSPLIT trajectories for advection). It is scoped to support extending Richardson et al. (2024), who showed that AIRS-derived CAPE — advected forward with forecast winds — skillfully predicts heavy hourly precipitation over the **central-eastern U.S. / Great Plains**, toward an improved extreme-convection signal that also incorporates SMAP.

Scope constraints (per study design).

- **Observable from AIRS:** temperature and moisture profiles → CAPE, CIN, lapse rates, mid-level EML/lid strength, column/lower-tropospheric humidity (PWAT), LCL/LFC, and (with advection) their time tendency.
- **Observable from SMAP:** surface (and derived root-zone) soil moisture — point values, spatial gradients/heterogeneity, and antecedent-memory time series.
- **Region:** Eastern U.S. and the Great Plains specifically (not the Sahel/semi-arid tropics where much of the mechanistic literature was developed). Where the strongest evidence comes from other regions, it is retained as a *mechanism analog* and paired with the U.S. counterpart.

How to read it. Mechanisms are grouped into categories (A–M). Each entry states the physical mechanism, the sign/nature of its effect on CI, its key references, and — most importantly — an **AIRS/SMAP observability + regional-relevance tag**:

-  **In scope** — directly derivable from AIRS and/or SMAP and relevant to Eastern/Great Plains US.
-  **Partial** — the *environmental preconditioning* is AIRS/SMAP-observable, but the km-scale trigger itself is not.
-  **Out of scope** — sub-grid/transient or requires data we don't have; listed for completeness.

Section N gives the prioritized shortlist keyed strictly to AIRS + SMAP over the target region.

Conceptual framing: the ingredients-based view

Deep moist convection requires the near-simultaneous presence of three ingredients: **moisture**, **conditional instability**, and a **lifting mechanism** sufficient to raise a parcel to its level of free convection (LFC), i.e., to overcome convective inhibition (CIN). Heavy precipitation additionally requires high **precipitation efficiency** and slow system motion (duration). This "ingredients-based" decomposition is the standard organizing principle and is worth adopting as the review's backbone.

- Doswell, Brooks & Maddox (1996) — canonical ingredients-based methodology for heavy-precipitation/flash-flood forecasting.
- Johns & Doswell (1992) — ingredients framing applied to severe-storm forecasting (parameters: instability, moisture, lift, shear).
- Markowski & Richardson (2010) — graduate textbook synthesizing CI, boundaries, and mesoscale forcing.

- Weckwerth & Parsons (2006) — the definitive observational review of CI mechanisms (IHOP_2002); emphasizes that CI is overwhelmingly boundary-driven and that moisture/lift at the km scale is the hardest and most decisive ingredient to observe.
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A. Conditional/thermodynamic instability (CAPE, CIN)

Mechanism. Positive buoyancy integrated from the LFC to the equilibrium level (CAPE) sets the potential updraft strength; CIN is the energy barrier a parcel must overcome. This is the thermodynamic core of the Richardson et al. (2024) approach. CAPE/CIN depend sensitively on the low-level temperature/moisture and the mid-level lapse rate, all of which are modified by advection between an AIRS overpass and CI.

- Richardson, Kahn & Kalmus (2024) — AIRS T/q soundings + forecast-wind advection produce a CAPE field that predicts intense hourly precipitation over the central-eastern U.S., near ERA5 skill and far above raw AIRS soundings; demonstrates that mesoscale advection of the thermodynamic environment is a first-order control. **(Target paper.)**
- Moncrieff & Miller (1976) — classic treatment linking CAPE/buoyancy to convective structure.
- **Retrievability:** High. CAPE/CIN from AIRS, ERA5, RTMA; the paper's HYSPLIT/advection step directly addresses the time-evolution problem.

B. Low-level moisture and moisture (flux) convergence

Mechanism. Boundary-layer moisture largely determines the LFC/LCL and the amount of CAPE realized; local moistening lowers the barrier to CI. Horizontal convergence of moisture concentrates water vapor and forces low-level ascent along mesoscale lines. Moisture is the ingredient most poorly observed at the scales relevant to CI (Weckwerth & Parsons 2006).

- Banacos & Schultz (2005) — history and operational use of moisture flux convergence for CI; argues mass convergence is the physically appropriate diagnostic within an ingredients framework.
- Weckwerth (2000) — boundary-layer water-vapor variability strongly modulates whether CI occurs along a given boundary.
- **Retrievability:** Moderate-high. Column/lower-tropospheric moisture from AIRS/SMAP-adjacent products, PWAT and moisture convergence from reanalysis; sub-grid BL moisture heterogeneity remains hard.

C. Capping inversion, elevated mixed layer (EML), and the "lid"

Mechanism. A stable layer (often an EML advected from elevated terrain such as the Mexican Plateau) caps the boundary layer, suppressing premature convection and allowing CAPE to build until strong enough forcing or heating breaches the cap — frequently yielding explosive CI. Too strong a cap gives a "null" case; too weak allows early, non-severe convection.

- Carlson, Benjamin, Forbes & Li (1983) — conceptual model of EMLs in the severe-storm environment; the lid mechanism.
- Ribeiro & Bosart (2018) — climatology of EMLs over CONUS/N. America and their link to severe environments.
- **Retrievability:** High. Lapse-rate/CIN structure and lid strength from AIRS/ERA5 profiles.

D. Mesoscale boundaries and low-level convergence lines (dominant CI trigger)

Mechanism. The great majority of observed CI occurs at boundary-layer convergence lines, where forced ascent lifts parcels to the LFC. Boundaries include fronts, drylines, sea/lake breezes, thunderstorm outflow (gust fronts/cold pools), and their intersections. Colliding boundaries are especially productive.

- Wilson & Schreiber (1986) — foundational: 79% of storms (95% of intense storms) initiated near radar-observed convergence lines; colliding lines triggered CI in 71% of cases.
- **Drylines:** Duda & Gallus (2013)/Peterson-style climatologies; drylines separate moist Gulf air from dry continental air and are a preferred CI locus in the southern Great Plains.
- **Sea/lake breezes:** Wakimoto & Atkins (1994) and CI ahead of the sea-breeze front — thermally forced convergence boundaries.
- **Outflow boundaries / cold pools (secondary CI):** Purdom (1976, and Purdom & Marcus 1982) — satellite "arc clouds"; storm outflow triggers new/organized convection. Torri, Kuang & Tian (2015) — mechanical (gust-front vorticity) vs. thermodynamic (moisture accumulation at the edge) triggering by cold pools.
- **Retrievability:** Mixed. Fronts/drylines/sea-breezes partly resolvable in high-res reanalysis/satellite; individual gust fronts and km-scale collisions are largely sub-grid and a known predictability limit.

E. Boundary-layer structure and lifting to the LFC (thermals, rolls, PBL depth)

Mechanism. Even without a synoptic boundary, the daytime convective boundary layer produces coherent updrafts — thermals and horizontal convective rolls (HCRs) — whose intersections with boundaries locally enhance moisture and lift, controlling exactly where along a boundary CI occurs. PBL growth toward the LCL/LFC and entrainment drying are decisive.

- Weckwerth, Wilson, Wakimoto & Crook (1997) — environmental conditions for HCRs; roll updrafts modulate BL moisture and CI location.
- Weckwerth & Parsons (2006) — synthesis of BL controls on CI location/timing.
- **Retrievability:** Low for individual thermals/rolls (sub-grid); PBL depth, LCL, and entrainment tendencies are obtainable from reanalysis/soundings.

F. Synoptic-scale forcing and large-scale ascent

Mechanism. Upper-level troughs, jet-streak circulations, and quasi-geostrophic ascent provide broad, sustained lift and lapse-rate steepening that precondition the environment (reduce CIN, moisten, cool aloft) — setting the stage for mesoscale triggers. Large-scale ascent also advects EMLs and moisture into place.

- Doswell, Brooks & Maddox (1996); Johns & Doswell (1992) — synoptic ingredients and pattern recognition.
- Maddox (1980) — mesoscale convective complexes and their synoptic settings.
- **Retrievability:** High. QG forcing, jet-level divergence, 500-hPa vorticity advection, omega from reanalysis.

G. Low-level jet and nocturnal/elevated convection

Mechanism. The Great Plains nocturnal low-level jet (LLJ) transports moisture and warm-air advection northward, generating low-level convergence/isentropic ascent at its terminus (nose) and destabilizing elevated layers. It underlies the nocturnal warm-season precipitation maximum, often via elevated CI decoupled from the surface.

- Higgins et al. (1997) — LLJ control on summertime moisture transport and nocturnal precipitation over the central U.S.
- Geerts et al. (2017) — PECAN field campaign overview of nocturnal, elevated convection mechanisms.
- **Retrievability:** High for the LLJ itself (reanalysis winds; also the natural input for HYSPLIT trajectories); elevated LFC/CIN from profiles.

H. Gravity waves and atmospheric bores

Mechanism. In the stable nocturnal boundary layer, undular bores and trapped gravity waves — often generated by MCS cold pools impinging on the stable layer — lift convectively unstable air above the inversion, initiating or maintaining elevated convection.

- Parsons et al. (2019) — potential role of bores/gravity waves in initiating and maintaining nocturnal convection over the southern Great Plains (PECAN).
- Koch et al. — structure/evolution of convectively generated undular bores as CI agents.
- **Retrievability:** Low. Bores/waves are transient and sub-grid; generally out of scope observationally, but worth flagging for nocturnal cases.

I. Orographic and terrain-driven forcing

Mechanism. Terrain forces CI via mechanical uplift (flow forced over/around a barrier) and thermal forcing (daytime upslope/valley winds converging over ridges). Convection initiates where terrain-induced ascent locally overcomes CIN in a conditionally unstable, moist airmass.

- Houze (2012) — comprehensive review of orographic effects on precipitating clouds.
- Kirshbaum et al. (2018) — mechanisms of moist orographic convection and links to surface fluxes.
- **Retrievability:** High for terrain and thermally forced upslope flow (static + reanalysis winds); exact CI cells remain sub-grid.

J. Land surface and soil moisture (core of the proposed study)

Soil moisture modulates the partition of net radiation into latent vs. sensible heat (Bowen ratio), thereby controlling BL growth, humidity, temperature, and the LCL/LFC — with competing pathways whose net sign depends on regime and scale. This is the richest and most contested category, so it is broken out below.

J1. Temporal (one-dimensional) feedback — sign is regime-dependent.

- **Wet-soil / positive pathway:** higher latent heat flux moistens and cools the BL, lowers the LCL, and can raise CAPE / reduce CIN, favoring CI locally.
- **Dry-soil / negative pathway:** higher sensible heat flux deepens the BL, promotes stronger thermals that can break the cap, and (via a drier-but-hotter BL) can still reach the LFC; several observational studies find afternoon rain preferentially over locally dry soils.
- Tuttle & Salvucci (2016) — remotely-sensed causal analysis: **positive** feedback in the western U.S., **negative** in the eastern U.S.; regional aridity governs the sign. (**Uploaded paper; central to the study.**)

- Taylor et al. (2012) — across six continents, afternoon rain falls preferentially over locally drier soils; models tend to show the opposite (spurious positive feedback).
- Findell & Eltahir (2003a,b) — CTP–Hllow framework: early-morning lapse rate (CTP) and low-level humidity index (Hllow) classify soundings as wet-soil-advantage, dry-soil-advantage, or insensitive. Directly relevant to combining SMAP with AIRS soundings.
- Ek & Holtslag (2004) — mechanistic BL model showing how soil moisture, entrainment, and BL growth jointly determine whether cloud/CI is favored over wet vs. dry soils.
- Liu et al. (2022) — recent review of soil moisture's influence on convective activity (useful for structuring this category).

J2. Spatial heterogeneity — mesoscale circulations from soil-moisture gradients.

- Soil-moisture (and induced flux) gradients on ~10–50 km scales set up thermally direct mesoscale circulations (sea-breeze-like), whose convergence branch preferentially triggers CI on the dry side / downwind of gradients.
- Taylor et al. (2007) — observational case of soil-moisture-induced mesoscale circulations.
- Taylor et al. (2011) — Sahelian storm-initiation frequency doubled over ~30-km soil-moisture patterns via enhanced convergence.
- Guillod et al. (2015) — reconciles the debate: **spatial** correlation (rain over locally dry patches) and **temporal** correlation (rain on overall-wet, heterogeneous days) have opposite signs and coexist; the distinction matters for how SMAP fields are used.
- Klein & Taylor (2020) — dry soils (patches ≥ 200 km) can intensify propagating MCS cores downstream.

J3. Regional-aridity dependence and direct CI observations.

- AlNasser, Short Gianotti & Entekhabi (2026) — tracks CI directly via geostationary cloud-top temperature (not precipitation proxy) with SMAP soil moisture across North America, Africa, Australia (2015–2020); the wet- vs. dry-soil preference varies systematically with regional aridity. **(Uploaded paper; methodological template for direct CI + SMAP.)**

J4. Coupling frameworks and metrics (how to quantify L–A coupling).

- Koster et al. (2004) — GLACE: identifies "hotspots" of strong soil-moisture–precipitation coupling in semi-arid transition zones.
- Santanello et al. (2018) — the LoCo (Local Land–Atmosphere Coupling) process chain and its metrics (CTP–Hllow, mixing diagrams, heated-condensation framework, two-legged metrics) — a menu of diagnostics for linking SMAP to CI.
- Dirmeyer (2011) — the "two-legged" coupling metric (terrestrial + atmospheric legs).
- **Retrievability:** High for the soil-moisture state (SMAP) and derived flux/BL diagnostics from reanalysis; spatial-gradient metrics require careful scale choices.

K. Vertical wind shear (organization/mode and CI feedback)

Mechanism. Shear is not primarily a trigger but governs storm organization, longevity, and secondary CI: it interacts with cold-pool vorticity (RKW-type balance) to produce upright, sustained updrafts at gust fronts, enabling continued initiation. Relevant to whether initiated cells become long-lived, heavy-rain-producing systems.

- Weisman & Klemp (1982) — dependence of simulated storm mode (cells → multicell → supercell) on shear and buoyancy.
- Rotunno, Klemp & Weisman (1988) — RKW theory: optimal cold-pool/shear balance for squall-line maintenance and gust-front CI.
- **Retrievability:** High. Bulk shear (0–6 km), storm-relative helicity from reanalysis winds.

L. Aerosols and cloud microphysics

Mechanism. Aerosols acting as CCN alter droplet spectra, can delay warm rain, and may "invigorate" deep convection through enhanced latent heating aloft (freezing of extra condensate). Effects on precipitation are regime-dependent (can enhance or suppress), and are secondary to dynamics/thermodynamics for CI timing/location.

- Rosenfeld et al. (2008) — "Flood or drought": framework for aerosol effects on precipitation, including invigoration.
- Tao, Chen, Li, Wang & Zhang (2012) — review of aerosol impacts on convective clouds and precipitation.
- **Retrievability:** Moderate (AOD from satellite) but mechanistically noisy; likely out of scope for a first-order CI signal.

M. Antecedent conditions and land-surface memory

Mechanism. Prior precipitation sets soil-moisture memory (days–weeks), which conditions subsequent flux partitioning and thus later CI — a slow, persistent control that is well suited to SMAP's strengths and to conditioning composites.

- Koster et al. (2004); Santanello et al. (2018) — memory as the mechanism translating a slowly varying land state into an atmospheric influence.
- Tuttle & Salvucci (2017) — cautions on confounders (persistence, synoptic co-variability) when inferring causal soil-moisture→precipitation links; directly relevant to designing the study's statistical tests.
- **Retrievability:** High. SMAP time series; antecedent precipitation from IMERG/gauge/reanalysis.

Cross-cutting notes for the study design

1. **CAPE is necessary but not sufficient.** Richardson et al. (2024) show advected CAPE already captures much skill for heavy rain in the central-eastern U.S., but CI *timing and location* are set by lift (category D/E) and CIN release (C) — the parts least visible to a polar-orbiter snapshot. SMAP's value proposition is that it constrains the *surface-flux* → *BL* → *CIN/LCL* chain that governs whether CAPE is realized.
2. **Sign of the soil-moisture effect is not universal** (J1–J3): it flips with aridity (Tuttle & Salvucci 2016; AlNasser et al. 2026) and with whether you measure spatial vs. temporal correlations (Guillod et al. 2015). Any SMAP predictor should be regime-stratified.
3. **Spatial gradients may matter more than point values** (Taylor et al. 2007, 2011; Klein & Taylor 2020) — consider SMAP gradient/heterogeneity features, not just local soil moisture.
4. **Confounding is the central inferential risk** (Tuttle & Salvucci 2017): soil moisture, antecedent rain, and synoptic forcing co-vary. HYSPLIT back-trajectories can help attribute the airmass and separate advective from local-surface contributions.

5. **Regional specificity.** Over the **Eastern U.S.**, the temporal soil-moisture–precipitation feedback is observationally **negative** (Tuttle & Salvucci 2016) — wetter-than-normal soils are associated with *reduced* subsequent rain probability — whereas the West is positive. Over the **Great Plains**, CI is dominated by drylines, the nocturnal LLJ, EML/lid release, and nocturnal elevated MCSs; the region is also a GLACE-type coupling "hotspot" and hosts the strongest afternoon soil-moisture signals in CONUS (Findell & Eltahir 2003b). A single SMAP predictor should therefore be **regime/region-stratified**, not applied with one sign.

N. Prioritized shortlist: mechanisms observable from AIRS + SMAP over the Eastern/Great Plains US

This is the working subset that satisfies all three constraints (AIRS-observable and/or SMAP-observable, and regionally relevant). It is the recommended predictor menu.

#	Mechanism / predictor	Primary source	Category	Key refs
1	CAPE / CIN (and advected tendency)	AIRS profiles + HYSPLIT	A	Richardson et al. 2024
2	Mid-level lapse rate / EML "lid" strength	AIRS profiles	C	Carlson et al. 1983; Ribeiro & Bosart 2018
3	Low-level moisture / PWAT / LCL-LFC	AIRS profiles	B	Weckwerth 2000; Banacos & Schultz 2005
4	CTP + Hllow (early-day sounding classifier of wet- vs dry-soil advantage)	AIRS profiles	J1	Findell & Eltahir 2003a,b
5	SMAP soil moisture (point value) — regime-stratified (E-US negative, Plains mixed)	SMAP	J1/J3	Tuttle & Salvucci 2016; AlNasser et al. 2026
6	SMAP spatial gradient / heterogeneity (mesoscale-circulation proxy)	SMAP	J2	Taylor et al. 2007, 2011; Klein & Taylor 2020
7	Antecedent soil-moisture memory (days–weeks)	SMAP time series	M	Koster et al. 2004; Tuttle & Salvucci 2017
8	Land–atmosphere coupling metrics (mixing diagram, HCF, two-legged) combining AIRS + SMAP	AIRS + SMAP	J4	Santanello et al. 2018; Dirmeyer 2011

Explicitly deprioritized (flagged above as 🟡 / ⚡): individual gust fronts/cold-pool collisions, horizontal convective rolls, thermals, bores/gravity waves, aerosols, and — as *direct* triggers — drylines/LLJ/orography (their thermodynamic *preconditioning* is captured via CAPE/CIN/lapse-rate, but the km-scale lift is not resolved by AIRS/SMAP). HYSPLIT is the tool for the advective/airmass-history piece the snapshot instruments miss.

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(and especially at collisions of) convergence lines.

Prepared as a scoping review to structure predictor selection for an AIRS-CAPE + SMAP extreme-convection study. Some categories (H, L, and sub-grid parts of D/E) are flagged as likely out of observational scope; the next step is to intersect this list with the specific AIRS/SMAP/HYSPLIT/reanalysis fields actually available.